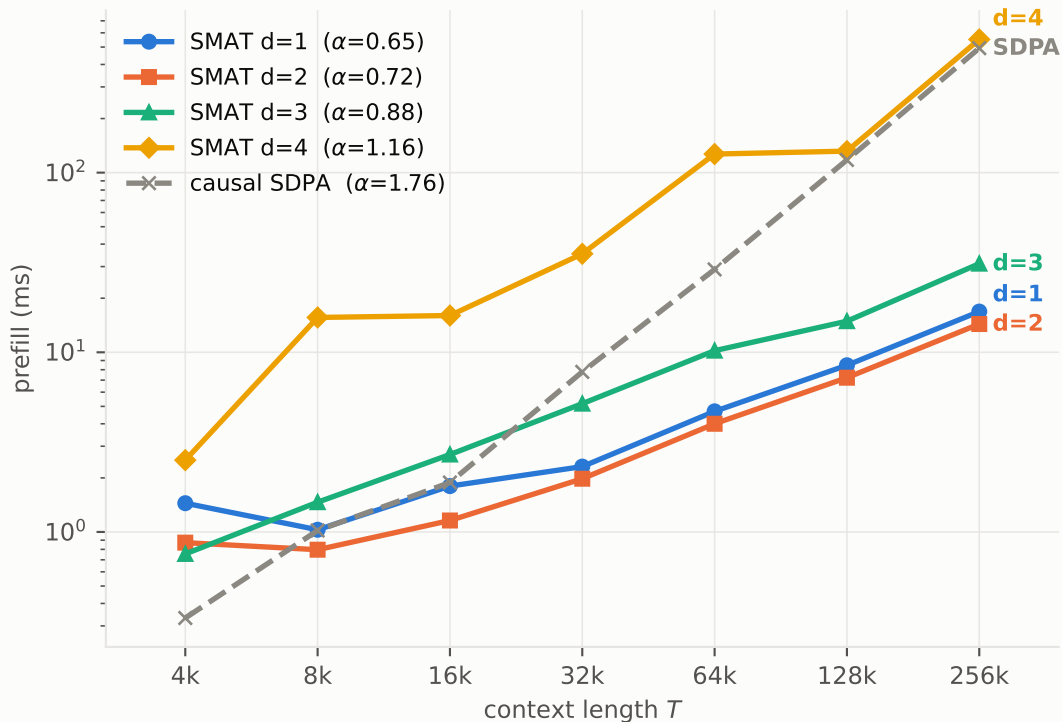


Prefill cost against causal attention



Total prefill is a $\Theta(T)$ scan plus a $\Theta(T^{2-3/d})$ long-range term, so $\alpha \approx 1$ for $d \leq 3$; fig2 isolates the long-range term.